

Verified Results on Nonboundary Localization Quotients

PBW structure, simplicity, the maximal submodule, and Takiff realizations

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Abstract

This note collects the audited results on nonboundary localization quotients of smooth induced modules over affine $\mathfrak{sl}_2$. In the deep region, under the simplicity assumption on the original module, there is an exact simplicity criterion; at integral parameters the locally nilpotent part of the opposite root is the unique maximal submodule. The basic example is induced from a Takiff localization quotient, but the whole affine module does not factor through any finite-order current-algebra truncation. The internal submodule classification of G_c and shallow-region simplicity are not among the proved statements. The main sources are the unified note and the addendum [2, 3]; the erroneous weight-tail filtration of an earlier structure draft is not used here.

1 Setup and parameter ranges

All objects are over $\mathbb{C}$. Omitting first the degree derivation, let

$$\mathfrak{g} = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K, \quad U = U(\mathfrak{g}), \quad x_i = x \otimes t^i,$$

with

$$\begin{aligned} [e_i, f_j] &= h_{i+j} + i\delta_{i+j,0}K, & [h_i, h_j] &= 2i\delta_{i+j,0}K, \\ [h_i, e_j] &= 2e_{i+j}, & [h_i, f_j] &= -2f_{i+j}, \end{aligned}$$

K central and $[e_i, e_j] = [f_i, f_j] = 0$. Fix $a, b \in \mathbb{Z}$ with $L = a + b + 1 \geq 1$ and set

$$S_{a,b} = \text{span}\{K, h_i \ (i \geq 0), e_i \ (i > a), f_j \ (j > b)\}.$$

The character χ satisfies $\chi(K) = \kappa$ and $\chi(h_i) = \lambda_i$, vanishes on root vectors, and we set $\lambda_i = 0$ for all $i > L$. Write

$$M = M_{a,b}(\chi) = U \otimes_{U(S_{a,b})} \mathbb{C}_\chi, \quad u = 1 \otimes 1.$$

The FGXZ criterion is [1, Theorem 1.1]

$$M \text{ simple} \iff \lambda_L \neq 0 \text{ and } \kappa \neq -2. \tag{1}$$

Throughout we fix a nonboundary index $c < b$ and write

$$\begin{aligned} F &= f_c, & d_* &= b - c, & D_F M &= U[F^{-1}] \otimes_U M, \\ Q_c &= \mathcal{T}_F M = D_F M / M, & w_m &= F^{-m}u + M \quad (m \geq 1). \end{aligned}$$

$\text{ad } F$ is locally nilpotent, so the Ore localization exists, and PBW makes F injective on M . Q_c is a quotient localization, not the full localization; F^{-1} is not defined on Q_c . We distinguish

$$\begin{aligned} \text{deep region:} & \quad c \leq -a - 1, \quad d_* \geq L, \\ \text{shallow region:} & \quad -a \leq c < b, \quad 1 \leq d_* < L. \end{aligned}$$

Unless stated otherwise, the structural results do not require (1).

2 Structure at arbitrary nonboundary indices

Proposition 2.1 (PBW, root modes, and weight spaces). Let $\mathcal{X}$ be the set of ordered PBW monomials (including the empty word) in the free modes

$$f_j \ (j \leq b, j \neq c), \quad e_i \ (i \leq a), \quad h_r \ (r < 0).$$

Then $\{F^{-m}Xu + M : m \geq 1, X \in \mathcal{X}\}$ is a basis of Q_c , and

$$\ker(F^r|_{Q_c}) = \text{span}\{F^{-m}Xu + M : 1 \leq m \leq r, X \in \mathcal{X}\}.$$

In particular $Q_c \neq 0$, and every nonzero submodule meets $\ker F$ nontrivially. The negative root modes act as follows:

index	action on Q_c
$j = c$	surjective and locally nilpotent
$j \leq b, j \neq c$	injective but not surjective
$j > b$	locally nilpotent

Q_c is a smooth weight module with $K = \kappa$ and h_0 semisimple, and

$$\text{Supp}_{h_0} Q_c = \lambda_0 + 2\mathbb{Z}, \quad \dim(Q_c)_\eta = \aleph_0 \quad (\eta \in \lambda_0 + 2\mathbb{Z}).$$

It has no nonzero finite-dimensional submodules or quotients, and it is not integrable for the full affine algebra.

Proof. Put F first among the free PBW variables: M is a free $\mathbb{C}[F]$ -module, and localization plus quotienting gives the basis. The remaining free f_j are independent polynomial variables. For the tail root modes use $f_j u = 0$ and the local nilpotence of $\text{ad } f_j$. The weight formula is $h_0(F^{-m}Xu) = (\lambda_0 + 2m + 2\#e(X) - 2\#f(X))F^{-m}Xu$; powers of h_{-1} give infinite multiplicities. Localization preserves smoothness, see (2). On a finite-dimensional quotient F cannot be both surjective and locally nilpotent; a finite-dimensional smooth affine module is trivial, and the injectivity of f_{c+1} excludes trivial submodules. $\square$

Proposition 2.2 (Cartan modes, the gap, and Hom-vanishing). If $1 \leq i \leq d_*$, $0 \neq v \in Q_c$ and $z \in \mathbb{C}$, then

$$\text{pole}_F((h_i - z)v) = \text{pole}_F(v) + 1.$$

Hence h_i has no nonzero locally finite vectors; if $i > d_*$, then $h_i - \lambda_i$ is locally nilpotent on Q_c . The set of locally nilpotent negative root modes is exactly $\{c\} \cup \{j : j > b\}$. Consequently Q_c is not isomorphic to any standard module $M_{a',b'}(\psi)$, and

$$\text{Hom}_{\mathfrak{g}}(M_{a',b'}(\psi), Q_c) = 0, \quad \text{Hom}_{\mathfrak{g}}(\mathcal{T}_{f_c}M, \mathcal{T}_{f_{c'}}M) = 0 \quad (c \neq c', c, c' \leq b).$$

Proof. In the full localization,

$$\begin{aligned} h_i F^{-m} &= F^{-m} h_i + 2m F^{-m-1} f_{c+i}, \\ e_i F^{-m} &= F^{-m} e_i - m F^{-m-1} (h_{i+c} + i \delta_{i+c,0} K) \\ &\quad - m(m+1) F^{-m-2} f_{i+2c}. \end{aligned} \tag{2}$$

For $i \leq d_*$ the top-pole coefficient contains the nonzero free variable f_{c+i} ; for $i > d_*$ one has $h_i w_m = \lambda_i w_m$, and local nilpotence follows from smoothness and positive index shifts. Standard modules have a full upper tail of locally nilpotent f -modes and a cyclic vector that is an h_1 -eigenvector. The last Hom-vanishing comes from f_c -torsion in the source and f_c -injectivity in the target. Here $Q_c = \sum_{m \geq 1} U w_m$: grade numerator words by $2\#e + \#h$; moving words past denominators gives correction terms of strictly lower grade, so induction applies. $\square$

3 Deep region: simplicity and the unique maximal submodule

In this section $c \leq -a - 1$, and we set

$$E = e_{-c}, \quad H = h_0 - cK, \quad \mu = \lambda_0 - c\kappa, \quad G_c = \{v \in Q_c : E^n v = 0 \text{ for some } n \geq 1\}.$$

Then (E, H, F) is an $\mathfrak{sl}_2$ -triple with $Eu = 0$, $Hu = \mu u$, and

$$Ew_m = -m(\mu + m + 1)w_{m+1}, \quad Hw_m = (\mu + 2m)w_m. \quad (3)$$

Proposition 3.1 (No simplicity assumption on the original module). G_c is a proper submodule of Q_c , and

$$G_c = 0 \iff \mu \notin \mathbb{Z}.$$

If $\mu \in \mathbb{Z}$, then for $k \geq \max(1, \mu + 2)$,

$$0 \neq f_b^k w_1 \in G_c, \quad E^{2k-\mu-1}(f_b^k w_1) = 0.$$

E is injective on Q_c/G_c . Moreover Q_c is always cyclic, and

$$Q_c = Uw_1 \iff \mu \notin \{-2, -3, \dots\}; \quad \mu = -s, s \geq 2 \implies Q_c = Uw_s \neq Uw_1.$$

Proof. ad E is locally nilpotent, so G_c is a submodule; (3) for sufficiently large m produces vectors outside G_c . If $0 \neq v \in \ker E$ is an H -weight vector of weight ν , let n be minimal with $F^n v = 0$; then $0 = EF^n v = n(\nu - n + 1)F^{n-1}v$, so $\nu = n - 1 \in \mathbb{Z}$. With $\nu \in \mu + 2\mathbb{Z}$ this gives $G_c = 0$ in the nonintegral case.

In the integral case let $Y = f_b$ and $\eta = \lambda_{d*}$. On $\mathbb{C}[F, Y]u$,

$$E(F^j Y^\ell u) = j(\mu - j + 1 - 2\ell)F^{j-1}Y^\ell u + \ell\eta F^j Y^{\ell-1}u.$$

Take $v_k = \sum_{j=0}^k a_j F^j Y^{k-j} u$ with

$$a_0 = 1, \quad a_j = -\frac{(k-j+1)\eta}{j(\mu-2k+j+1)}a_{j-1}.$$

The denominators are nonzero, and $Ev_k = 0$, $Hv_k = (\mu - 2k)v_k$. Applying the rank-one formula to $F^{-1}v_k$ and passing to the quotient gives the claimed nonzero vector and the nilpotence bound. $Ev \in G_c \implies v \in G_c$ gives injectivity on the quotient. Cyclicity follows from (3), $Fw_m = w_{m-1}$ ($w_0 = 0$) and generation by pure fractions; for $\mu = -s$ one has $w_1 \in G_c$ but $w_s \notin G_c$. $\square$

Theorem 3.2 (Exact deep-region criterion and simple head). Assume

$$\lambda_L \neq 0, \quad \kappa \neq -2, \quad c < b, \quad c \leq -a - 1. \quad (4)$$

Then

$$\boxed{Q_c \text{ simple} \iff \lambda_0 - c\kappa \notin \mathbb{Z}.}$$

For every μ , every proper submodule is contained in G_c . Hence G_c is the unique maximal submodule, $\text{rad } Q_c = G_c$, Q_c/G_c is simple, and Q_c is indecomposable. If $\mu \in \mathbb{Z}$, then $0 \neq G_c \neq Q_c$ and there is a nonsplit exact sequence

$$0 \longrightarrow G_c \longrightarrow Q_c \longrightarrow Q_c/G_c \longrightarrow 0. \quad (5)$$

In this integral case $\text{soc } Q_c = \text{soc } G_c$, which does not determine whether that socle is zero.

Proof. By (1), M is simple; E is locally nilpotent on M , F is injective, and H is semisimple. We prove the key containment [3, Section 2]. Let $N \subseteq Q_c$ be a submodule not contained in G_c , and let $P \subseteq D_F M$ be its inverse image. Choose an H -weight vector $w \in P$ whose class $q \in N$ satisfies $E^r q \neq 0$ for all $r \geq 0$. The rank-one Casimir

$$C = H^2 + 2H + 4FE$$

acts locally finitely on M and $D_F M$, because for $H z = \eta z$,

$$F^n E^n z = A_n(C; \eta) z, \quad A_n(t; \eta) = 4^{-n} \prod_{j=0}^{n-1} (t - (\eta + 2j)(\eta + 2j + 2)).$$

E is locally nilpotent on M , and C commutes with F^{-1} . Choose $0 \neq p(t)$ with $p(C)w = 0$. Replacing w by a sufficiently high $v = E^r w$ of weight ν , we may arrange $p((\nu + 2j)(\nu + 2j + 2)) \neq 0$ for all $j \geq 0$. A Bézout identity provides polynomials B_n with

$$v = F^n E^n B_n(C) v \in F^n P \quad (n \geq 0).$$

Thus $0 \neq v \in W(P) := \bigcap_{n \geq 0} F^n P$. The local nilpotence of $\text{ad } F$ makes $W(P)$ a U -submodule: if $(\text{ad } F)^{\ell+1} s = 0$, then

$$s F^{n+\ell} = \sum_{j=0}^{\ell} (-1)^j \binom{n+\ell}{j} F^{n+\ell-j} (\text{ad } F)^j(s).$$

$W(P)$ is also stable under F^{-1} and meets M nontrivially. Since M is simple, $M \subseteq W(P)$; localizing gives $P = D_F M$ and $N = Q_c$. Together with Proposition 3.1 this yields the criterion and the unique maximal submodule. In any nontrivial direct-sum decomposition both summands would be contained in G_c , so Q_c is indecomposable and (5) does not split. In the integral case every simple submodule is proper, hence the socle equality. $\square$

Remark 3.3 (Cyclicity is not simplicity). For $M_{-1,1}$, $c = 0$, $\lambda_0 = 0$, $\lambda_1 \neq 0$, $\kappa \neq -2$, one has $Q_0 = U w_1$, but $0 \neq f_1^2 w_1 \in G_0$ and $e_0^3 f_1^2 w_1 = 0$, so Q_0 is not simple. Theorem 3.2 must not be transferred verbatim to the raw critical or degenerate induced modules.

4 Current limits: the maximal submodule's interior and the shallow region

Proposition 4.1 (Transverse $\mathfrak{sl}_2$ structure; not an affine submodule classification). In the deep region let $\mathfrak{s} = \langle E, H, F \rangle \cong \mathfrak{sl}_2$ and $P_\nu = \ker E \cap \{v \in Q_c : H v = \nu v\}$. As $\mathfrak{s}$ -modules,

$$G_c \cong \bigoplus_{\nu \in \mathbb{Z}_{\geq 0}} P_\nu \otimes L(\nu),$$

where $L(\nu)$ is the finite-dimensional simple $\mathfrak{sl}_2$ -module of highest weight ν , and the multiplicity spaces P_ν need not be finite-dimensional. For $0 \neq p \in P_\nu$ one has $\nu \in \mu + 2\mathbb{Z}$ and $\text{pole}_F(p) = \nu + 1$.

Proof. E, F are locally nilpotent on G_c and H is semisimple; PBW places every vector in a finite-dimensional $\mathfrak{s}$ -submodule. Complete reducibility gives the decomposition. A highest-weight vector satisfies $F^{\nu+1} p = 0$ and $F^\nu p \neq 0$; the kernel formula of Proposition 2.1 gives the pole equality. $\square$

One must not treat $\bigoplus_{\nu \geq n} P_\nu \otimes L(\nu)$ as a $\mathfrak{g}$ -submodule automatically. Affine modes have components that lower the transverse highest weight; for instance, for $p \in P_\nu$ with $\nu \geq 2$,

$$f_j p - \frac{1}{\nu} F h_{j-c} p - \frac{1}{\nu(\nu+1)} F^2 e_{j-2c} p \in P_{\nu-2} \quad (j \neq c).$$

This is a direct $\mathfrak{sl}_2$ projection formula; it is not asserted that its value is always zero. Consequently, at integral parameters the questions whether G_c is simple, the value of $\text{soc } G_c$, and whether Q_c has finite composition length are not settled by the results above. In particular one may not claim $\text{soc } Q_c = 0$ or infinite length; length two is equivalent to simplicity of G_c , which is likewise undecided.

Proposition 4.2 (Shallow region). If $-a \leq c < b$, then $E = e_{-c}$ is injective on Q_c , hence $G_c = 0$. If in addition $\lambda_L \neq 0$, then $Q_c = U w_1$.

Proof. E is a free PBW mode. Grade numerators by $2\#e + \#h$: the top action of E is multiplication by a free variable, and commutation corrections have lower grade, so E is injective. Let $Y = e_{L-c}$; the shallow region gives $Yu = 0$, $[Y, F] = h_L$ and $f_{L+c}u = 0$, hence $Yw_m = -m\lambda_L w_{m+1}$, which yields cyclicity. $\square$

In the shallow region E is not locally nilpotent on the original module M , so the proof of Theorem 3.2 cannot deduce simplicity of Q_c from $G_c = 0$. Under $\lambda_L \neq 0$ the nonzero cyclicity guarantees the existence of a maximal submodule; its uniqueness and the exact simplicity condition for Q_c remain undetermined.

5 The precise relation to Takiff modules

Write

$$\mathfrak{t}_N = \mathfrak{sl}_2[t]/(\mathfrak{sl}_2 \otimes t^{N+1}\mathbb{C}[t]) \quad (N \geq 1), \quad \mathfrak{p} = \mathfrak{sl}_2[t] \oplus \mathbb{C}K.$$

$\mathfrak{t}_1$ is the usual Takiff algebra; $N > 1$ gives higher-order truncated current algebras. *Being a Takiff module, containing a Takiff subspace, and being induced from a Takiff module are three different statements.*

5.1 The basic example: exact model and induced realization

Take $Q_0 = \mathcal{T}_{f_0} M_{-1,1}$ and define

$$V = \text{span}\{f_1^k w_m : m \geq 1, k \geq 0\} \cong \mathbb{C}[x, x^{-1}, y]/\mathbb{C}[x, y], \quad f_1^k w_m \longleftrightarrow x^{-m} y^k.$$

Then V is stable under $\mathfrak{sl}_2[t]$, all e_i, f_i, h_i ($i \geq 2$) act by zero on it, so it is a $\mathfrak{t}_1$ -module. The exact action is [2, Section 6]

$$\begin{aligned} f_0 &= x, & f_1 &= y, \\ h_0 &= \lambda_0 - 2x\partial_x - 2y\partial_y, & h_1 &= \lambda_1 - 2y\partial_x, \\ e_0 &= \lambda_0\partial_x + \lambda_1\partial_y - x\partial_x^2 - 2y\partial_x\partial_y, & e_1 &= \lambda_1\partial_x - y\partial_x^2. \end{aligned} \tag{6}$$

These operators preserve the polynomial subspace being quotiented out, so the action is well defined. Endowing K with the scalar action κ , PBW also gives

$$Q_0 \cong U(\mathfrak{g}) \otimes_{U(\mathfrak{p})} V. \tag{7}$$

Indeed, the natural map $X \otimes v \mapsto Xv$ sends the PBW basis of $U(\mathfrak{sl}_2 \otimes t^{-1}\mathbb{C}[t^{-1}]) \otimes V$ to the right-normal localized PBW basis. V is not a $\mathfrak{g}$ -submodule: for instance $h_{-1}w_1 = F^{-1}h_{-1}u + 2F^{-2}f_{-1}u + M \notin V$. Thus the basic example is precisely an *affine module induced from a Takiff localization quotient*.

5.2 Realizations available directly at general indices

For $M_{-1,N}$ let $W_N = \mathbb{C}[f_0, \dots, f_N]u$. This is a $\mathfrak{t}_N$ -module with $M_{-1,N} \cong U(\mathfrak{g}) \otimes_{U(\mathfrak{p})} W_N$ [1, Lemma 2.2]. The same PBW comparison gives, for $0 \leq r < N$,

$$\mathcal{T}_{f_r} M_{-1,N} \cong U(\mathfrak{g}) \otimes_{U(\mathfrak{p})} V_{N,r}, \quad V_{N,r} := \mathcal{T}_{f_r} W_N \cong \frac{\mathbb{C}[z_0, \dots, z_N, z_r^{-1}]}{\mathbb{C}[z_0, \dots, z_N]}. \quad (8)$$

The right-hand fractions are vector-space models; the action on $V_{N,r}$ comes from localizing W_N and taking the quotient. The higher-mode ideal vanishes on W_N and is preserved by $\text{ad } f_r$, so it also vanishes on $V_{N,r}$. (8) requires neither noncritical nor nondegenerate hypotheses, and it does not by itself give simplicity of the induced module.

Using the twisted convention $x \cdot_{\text{new}} v = \sigma_k(x)v$ with

$$\sigma_k(e_i) = e_{i+k}, \quad \sigma_k(f_i) = f_{i-k}, \quad \sigma_k(h_i) = h_i + k\delta_{i,0}K, \quad \sigma_k(K) = K,$$

take $k = a + 1$: then

$$(Q_c)^{\sigma_{a+1}} \cong \mathcal{T}_{f_r} M_{-1,L}(\chi'), \quad r = c + a + 1, \quad \chi'(h_0) = \lambda_0 + (a + 1)\kappa,$$

with the remaining parameters unchanged. Hence for $-a - 1 \leq c < b$ the Takiff induced realization above applies directly; this covers the whole shallow region and the deep-region endpoint $c = -a - 1$. This is a twisted isomorphism, not an untwisted one.

For arbitrary deep-region indices, the alternative normalization is

$$(Q_c)^{\sigma_{-c}} \cong \mathcal{T}_{f_0} M_{A,B}(\chi''), \quad A = a + c \leq -1, \quad B = b - c, \quad \chi''(h_0) = \mu.$$

Its subspace

$$V_B = \text{span}\{f_0^{-m} f_1^{k_1} \cdots f_B^{k_B} u + M : m \geq 1, k_i \geq 0\}$$

is stable under $\mathfrak{sl}_2[t]$ and is annihilated by $\mathfrak{sl}_2 \otimes t^{B+1}\mathbb{C}[t]$, so it is a $\mathfrak{t}_B$ -module. This is a direct consequence of positive-mode reordering. If $A < -1$, this note does not extend the basic induction isomorphism (7) verbatim to this subspace.

5.3 No finite-order truncation carries the whole nonboundary quotient

For every Q_c and every $R \geq 1$,

$$(\mathfrak{sl}_2 \otimes t^R \mathbb{C}[t])Q_c \neq 0. \quad (9)$$

Otherwise h_R would vanish on all of Q_c ; then $[h_R, f_{c+1-R}] = -2f_{c+1}$ forces f_{c+1} to vanish as well, contradicting Proposition 2.1. So even for the natural restricted $\mathfrak{sl}_2[t]$ action, the whole module Q_c is not a Takiff module of any finite order. Smoothness only says that *each vector* has its own bound annihilating high modes; there is no uniform truncation order for the whole module. Being induced from a Takiff module also does not place the result in the original standard family $M_{a',b'}$.

6 Further localization, central reduction, and derived properties

None of the following presupposes the internal classification of G_c . Proofs: [2, Sections 8–10] and [3, Section 5].

Full localization and annihilators. In the ordinary enveloping algebra U ,

$$\text{Ann}_U M = \text{Ann}_U D_F M = \text{Ann}_U Q_c.$$

$0 \rightarrow M \rightarrow D_F M \rightarrow Q_c \rightarrow 0$ does not split, and

$$\text{Hom}_{\mathfrak{g}}(Q_c, M) = \text{Hom}_{\mathfrak{g}}(Q_c, D_F M) = \text{Hom}_{\mathfrak{g}}(M, Q_c) = 0.$$

If M is simple, then M is the essential and the unique simple submodule of $D_F M$, so $D_F M$ is indecomposable. This is a different exact sequence from (5).

Commuting quotient localizations. If $j \leq b$ and $j \neq c$, then

$$\mathcal{T}_{f_j} \mathcal{T}_{f_c} M \cong \frac{D_{f_c, f_j} M}{D_{f_c} M + D_{f_j} M} \cong \mathcal{T}_{f_c} \mathcal{T}_{f_j} M.$$

Quotient localization in any finite set of distinct free f -modes is independent of the order. If $\lambda_L \neq 0$, localizing once each of $f_c, \dots, f_b$ produces

$$M_{a+b-c+1, c-1}(\chi^{(b-c+1)}), \quad \chi^{(k)}(h_0) = \lambda_0 + 2k,$$

with the remaining parameters unchanged. If in addition $\kappa \neq -2$, the final module is simple; this does not imply simplicity of the intermediate quotients.

Derived functors. For any U -module V , define $\mathcal{T}_F V = \text{coker}(V \rightarrow D_F V)$. Then

$$\begin{aligned} \mathcal{T}_F V &= (U[F^{-1}]/U) \otimes_U V, \\ L_1 \mathcal{T}_F(V) &\cong \{v \in V : F^n v = 0 \text{ for some } n \geq 1\}, \quad L_j \mathcal{T}_F(V) = 0 \quad (j \geq 2). \end{aligned}$$

Hence $D_F Q_c = \mathcal{T}_F Q_c = 0$, while $L_1 \mathcal{T}_F(Q_c) \cong Q_c$.

Central reduction and the derivation. If a commutative algebra R acts on V by U -module endomorphisms and $I \subseteq R$ is an algebraic ideal, then

$$\mathcal{T}_F(V/IV) \cong (\mathcal{T}_F V)/I(\mathcal{T}_F V).$$

The Sugawara operators on smooth modules at the critical level satisfy this hypothesis; the isomorphism itself guarantees neither nonvanishing nor injectivity of F after reduction nor simplicity of the quotient. Only ideal multiples by finite algebraic sums are involved; no assertion is made about topological quotients of closed ideals.

For $\tilde{\mathfrak{g}} = \mathfrak{g} \rtimes \mathbb{C}d$ with $[d, x_i] = ix_i$, put $\mathcal{I}(V) = U(\tilde{\mathfrak{g}}) \otimes_U V$. Then

$$\mathcal{I}(\mathcal{T}_F V) \cong \mathcal{T}_F \mathcal{I}(V),$$

with $\mathcal{I}(V) \cong \mathbb{C}[d] \otimes V$ and the action $x_i(p(d) \otimes v) = p(d - i) \otimes x_i v$, which is not the trivial tensor action. Induction does not in general preserve simplicity, the unique maximal submodule, or indecomposability, so Theorem 3.2 does not transfer verbatim.

Scope. The PBW structure, the mode profiles, and the cyclicity of the nonboundary quotients, together with the deep-region simple head, are established. The internal structure of G_c at deep integral parameters, shallow-region simplicity, nonboundary simplicity after critical reduction, and the full parameter isomorphism classification still require independent arguments.

References

- [1] V. Futorny, X. Guo, Y. Xue and K. Zhao, *Smooth representations of affine Kac–Moody algebras*, arXiv:2404.03855v2, 2024. Theorem 1.1; Section 2.4; Lemma 2.2.
- [2] Project unified note, *Boundary and Nonboundary Localization Quotients of Smooth Affine Modules*, September 2026. `localization_quotients_unified_en.tex`. This note uses its audited structural conclusions; its open problems are partially resolved by the addendum below.
- [3] Project addendum, *Additional Progress on Localization Quotients: Deep-region simplicity, the unique maximal submodule, and compatibility with central reduction*, September 15, 2026. `localization_quotients_additional_progress_en.tex`.

Repository version checked: 5f766461cb0a01d171b136b55a64cd1f214886de. This note is a summary of results with the necessary proofs; it does not constitute a classification of the full internal structure.